Bolla Lokesh Reddy

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.29
Sri Chaitanya Junior College, Guntur	Class XII (Board of Intermediate Education, Andhra Pradesh)	2023	9.57%
Bhashyam Educational Institution, Guntur	Class X (Board of Secondary Education, Andhra Pradesh)	2021	10.00%

EXPERIENCE

Adobe -- Machine Learning Engineering Intern

- Developed a **6-stage email layout featurization pipeline** integrating headless Chromium rendering, Gemini-based semantic annotation, canonical layout generation, multimodal embeddings, clustering, and deterministic feature extraction.
- Combined **Playwright/Chromium DOM geometry with Gemini-based annotations** to recover structural regions, parent-child hierarchies, and spatial relationships from rendered email HTML.
- Designed and evaluated **four progressively abstracted canonical layout representations** using DINOv2 and Gemini embeddings to identify recurring email layout families.
- Engineered **22 deterministic structural features** capturing layout composition, header and image strategies, CTA placement, grid structure, and content composition; validated the pipeline across two independent datasets.

PROJECTS

Distributed Bitcoin Mining Framework

Course Project under Prof.Sathya Peri | *GitHub*

- Designed and implemented a distributed Bitcoin mining framework supporting **static multithreaded partitioning, dynamic work stealing, and distributed nonce search**.
- Developed dynamic work allocation across threads and processes to address **load imbalance** in fixed-partition mining and improve parallel execution.
- Built a fault-tolerant distributed architecture with **Raft-based leader election, checkpointing, and recovery**, and benchmarked scalability across varying thread/process counts and mining difficulty.

Fault-Tolerant Distributed Job Processing System | *Self Project* | *GitHub*

- Built a fault-tolerant asynchronous job processing platform with **worker pools, explicit job lifecycle management, persistent state transitions, retries, and dead-letter queues**.
- Implemented **at-least-once delivery** using consumer groups, stale-message reclamation, exponential-backoff retries, and hash-based idempotency for safe recovery from worker failures.
- Added **API-key authentication, per-client rate limiting, Prometheus metrics**, and a 16-test integration suite with failure-injection testing against live Redis and PostgreSQL instances.

RISC-V Assembler and Cache Simulator

Course Project under Prof.Rajesh Kedia | *GitHub1* | *GitHub2*

- Developed a **RISC-V assembler and simulator** supporting instruction encoding, decoding, execution, register management, and memory operations across RISC-V instruction formats.
- Integrated a configurable **D-cache simulator** supporting different cache organizations, replacement policies, write policies, and tracking of hit, miss, and access statistics.
- Built a **CLI-based interface** for program execution, debugging, breakpoints, cache configuration, and runtime inspection, with systematic testing across edge cases and varied configurations.

Virtual Memory Management in gemOS

Course Project under Prof. Sathya Peri

- Implemented virtual memory management in gemOS with **VMA operations, protection, lazy allocation, page-fault handling, and on-demand page-table updates**.
- Developed **Copy-on-Write fork** using shared physical frames, reference counting, and CoW fault handling.

COMPETITIVE PROGRAMMING

- *Codeforces* **:** 1500+ rating
- *LeetCode* **:** 500+ problems solved

SKILLS

- **Languages:** C, C++, Python, JavaScript, SQL
- **Backend & Systems:** Node.js, Express.js, FastAPI, REST APIs, Redis, PostgreSQL, MongoDB, MPI
- **Development & Tools:** Git, GitHub, Linux, CMake, Catch2

RELEVANT COURSES

- **Computer Science:** Data Structures, Algorithms, Operating Systems, Database Management Systems, Computer Architecture, Computer Networks, Distributed Computing, Compiler Design, Software Engineering, Natural Language Processing, Foundations of Machine Learning, Deep Learning
- **Mathematics:** Probability and Random Variables, Discrete Mathematics, Linear Algebra, Calculus

POSITIONS OF RESPONSIBILITY

Teaching Assistant - CSE IIT Hyderabad

- Served as TA for the course Operating Systems , Data Structures

Machine Learning Coordinator : Tinkerer's Lab, IIT Hyderabad

- Organizing hands-on ML workshops and mentoring peers in ML model development and deployment.

OCS Internship Coordinator : Office of Career Services, IIT Hyderabad

- Assisting in internship process coordination, outreach, and student-industry communication.